

Optimization

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Section 1

Definitions

Constrained choice

Optimization selects the *best feasible point*.

- ◆ The objective assigns a value to every feasible point.
- ◆ Constraints determine the feasible set.

Example: maximize a utility function over budget-feasible set of product bundles.

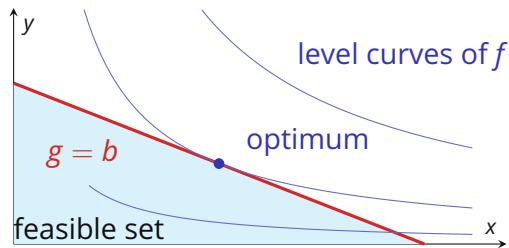


Figure 1 · Feasible set (shaded), level curves of the objective, and the best feasible point (optimum).

Optimization problems

General form. Given a feasible set $S \subseteq \mathbb{R}^n$ and an objective $f : S \rightarrow \mathbb{R}$,

$$\max_{x \in S} f(x).$$

Maximizer and maximum. In the set-builder and quantifier notation from Preliminaries,

$$\arg \max_{x \in S} f(x) := \{x^* \in S : \forall x \in S, f(x) \leq f(x^*)\}.$$

When this set is nonempty, the **maximum value** is $\max_{x \in S} f(x) = f(x^*)$ for every maximizer x^* .

Existence means $\arg \max_{x \in S} f(x) \neq \emptyset$; uniqueness means this set has one element.

Video: [Steve Brunton, *The anatomy of an optimization problem*](#).

Four questions in an optimization problem

Once the objective and feasible set are fixed, separate four logically different questions.

Existence

Is $\arg \max_{x \in S} f(x)$ nonempty?

Globality and uniqueness

When is a local optimum global, and when is the optimizer unique?

Optimality conditions

Which conditions are necessary, and when are they sufficient?

Sensitivity

How do the optimal choice and optimized value change with parameters?

Compactness and continuity address existence; FOC, SOC, Lagrange and KKT give optimality conditions; curvature connects local and global conclusions; comparative statics and the envelope theorem study parameter changes.

Consumer and firm problems

The same template separates three objects: the **choice variable**, the **feasible set**, and the **objective**.

Consumer choice

$$\max_{x \in \mathbb{R}_+^n} u(x) \quad \text{s.t.} \quad p \cdot x \leq w.$$

- ◆ x : consumption bundle.
- ◆ $p \cdot x \leq w$: budget feasibility.
- ◆ $u(x)$: objective to maximize.

Firm choice

$$\max_{q \geq 0} pq - c(q).$$

- ◆ q : output choice.
- ◆ $q \geq 0$: feasible production scale.
- ◆ $pq - c(q)$: profit objective.

From Convexity, the standard linear budget set is convex; when u represents the consumer's preferences, preference convexity is equivalent to quasiconcavity of u . Strict quasiconcavity makes an attained maximizer unique on the budget set.

Further reading: [Border, Preference and Demand Examples](#); [Border, Examples of Cost and Production Functions](#).

Data fitting and allocation

The same notation covers estimation and allocation; minimizing C is equivalent to maximizing $-C$.

Least squares fit

$$\min_{\beta \in \mathbb{R}^k} \|y - X\beta\|^2$$

- ◆ β : model coefficients.
- ◆ $\mathbb{R}^k$: unrestricted feasible set.
- ◆ $\|y - X\beta\|^2$: prediction error.

Resource allocation

$$\begin{aligned} \min_{a \in \mathbb{R}_+^m} C(a) \\ \text{s.t. } \sum_{i=1}^m a_i = R. \end{aligned}$$

- ◆ a_i : amount assigned to use i .
- ◆ $\sum_i a_i = R$: resource feasibility.
- ◆ $C(a)$: cost to minimize.

Local and global extrema

Let $x^* \in S$ and let $B_\epsilon^S(x^*) = S \cap B_\epsilon^{\mathbb{R}^n}(x^*)$, the open ball in S from Functions.

Global maximum $\forall x \in S, f(x) \leq f(x^*)$.

Local maximum $\exists \epsilon > 0 \forall x \in B_\epsilon^S(x^*), f(x) \leq f(x^*)$.

Global minimum $\forall x \in S, f(x) \geq f(x^*)$.

Local minimum $\exists \epsilon > 0 \forall x \in B_\epsilon^S(x^*), f(x) \geq f(x^*)$.

Global compares x^* with all of S ; local only with $B_\epsilon^S(x^*)$ for some $\epsilon > 0$. Since $B_\epsilon^S(x^*) \subseteq S$,

$$x^* \text{ global max} \Rightarrow x^* \text{ local max}, \quad x^* \text{ global min} \Rightarrow x^* \text{ local min}.$$

Hence a sufficient condition for a global extremum is sufficient for the corresponding local extremum, while a necessary condition for a local extremum is necessary for the corresponding global extremum. The converses need not hold.

Video: [Khan Academy](#), *Multivariable maxima and minima*.

Local vs. global extrema

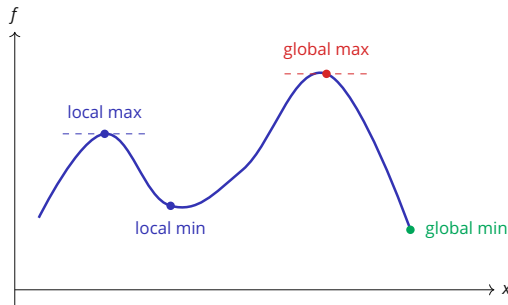


Figure 2 · Local and global extrema.

Local extrema compare nearby feasible points; a global extremum may occur at a boundary point.

Local optima are global

Convexity of the feasible set keeps every segment from a local candidate to another feasible point inside the feasible set.

Proposition. Let $S \subseteq \mathbb{R}^n$ be convex and let $f : S \rightarrow \mathbb{R}$.

- (i) If f is convex, every local minimizer is global. If f is strictly convex, f has **at most one** minimizer.
- (ii) If f is concave, every local maximizer is global. If f is strictly concave, f has **at most one** maximizer.

Further reading: MIT 6.253, *Convex Analysis and Optimization*, p. 59; Boyd and Vandenberghe, *Convex Optimization*, §4.2.

Video: Mohamed Omar, *Convex: Local vs Global Minimum*.

A constrained maximum

Maximize $f(x)$ over $x \in [-1, 2]$.

- ◆ The unconstrained maximizer can lie outside the feasible set.
- ◆ The constrained maximizer can then sit on the boundary.

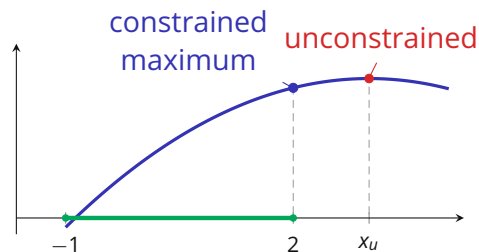


Figure 4 · The feasible interval is green.

The constrained maximum occurs at the boundary point $x = 2$.

Useful transformations

- ◆ Minimizers of f are maximizers of $-f$.
- ◆ If g is strictly increasing on the range of f , then composition preserves the optimizer set:

$$\arg \max_{x \in S} f(x) = \arg \max_{x \in S} (g \circ f)(x).$$

Log trick. For $f(x) = A \prod_i x_i^{\alpha_i} > 0$, $\ln f = \ln A + \sum_i \alpha_i \ln x_i$. Maximizing a positive likelihood $L(\theta)$ is likewise equivalent to maximizing $\ell(\theta) = \ln L(\theta)$.

This applies composition and monotonicity from Functions: $g \circ f$ preserves the ranking of feasible points when g is strictly increasing. The optimizer set is unchanged, but the numerical value of the objective changes.

Further reading: [Boyd, Ch. 20, §§20.3.3–20.3.5.](#)

Preliminaries applied: logic of an optimizer

Let $S \subseteq \mathbb{R}^n$, $f : S \rightarrow \mathbb{R}$, and $x^* \in S$.

- (a) Negate the statement $\forall x \in S, f(x^*) \geq f(x)$.
- (b) Let $I \subseteq \mathbb{R}$ be an interval containing every value of f and let $g : I \rightarrow \mathbb{R}$ be strictly increasing ($u < v$ in I implies $g(u) < g(v)$), say $I = (0, \infty)$ and $g = \ln$ for $f > 0$. Composition replaces the objective on the same S by

$$(g \circ f)(x) = g(f(x)), \quad x \in S.$$

Prove $\arg \max_{x \in S} f(x) = \arg \max_{x \in S} g(f(x))$.

- (c) The EVT is sufficient for existence, so violating its hypotheses leaves existence open. Show from the definition that $\max_{x \in \{x : x \geq 2, x \leq 1\}} f(x)$ and $\max_{x \in (0,1)} x$ have no maximizer, then give a problem that violates the same hypotheses and does have one.
- (d) Which assumptions in the EVT give existence, and what does strict concavity add on a convex feasible set?

Preliminaries applied: logic of an optimizer

(a) The negation is

$$\exists x \in S \text{ such that } f(x) > f(x^*).$$

- (b) Every value $f(x)$ lies in I , where strict monotonicity of g gives both $u \leq v \Rightarrow g(u) \leq g(v)$ and $u > v \Rightarrow g(u) > g(v)$. If $x^* \in \arg \max_S f$, then $f(x) \leq f(x^*)$ for every $x \in S$, so $(g \circ f)(x) = g(f(x)) \leq g(f(x^*)) = (g \circ f)(x^*)$ and x^* maximizes $g \circ f$. Conversely, if x^* maximizes $g \circ f$ and some $x \in S$ satisfied $f(x) > f(x^*)$, then $g(f(x)) > g(f(x^*))$, a contradiction; hence $f(x) \leq f(x^*)$ throughout S .
- (c) The first feasible set is empty, so it contains no maximizer. In the second, $(1+x)/2$ lies in $(0, 1)$ and exceeds x for every $x \in (0, 1)$, so no candidate survives; the supremum 1 sits outside $(0, 1)$. Violating the EVT hypotheses by itself gives neither conclusion: $\max_{x \in (0, 1)} x(1-x)$ has the same open feasible set and attains its maximum at $x = 1/2$.
- (d) A nonempty compact feasible set and a continuous objective give existence. Strict concavity gives at most one maximizer: two distinct maximizers would have a midpoint with a strictly larger objective value.

Maximizers, maximum values, and local extrema

Let $f(x) = x^3 - 3x$ on $S = [-2, 2]$.

- (a) Find $\arg \max_{x \in S} f(x)$ and $\max_{x \in S} f(x)$.
- (b) Classify $x = -1$, $x = 1$, and $x = 2$ as local or global maxima or minima relative to S .
- (c) Explain why the answer to part (a) is a set, whereas the maximum is a number.

Maximizers, maximum values, and local extrema

Since $f'(x) = 3(x^2 - 1)$, the interior critical points are $x = \pm 1$. Evaluating them and the endpoints gives

$$f(-2) = -2, \quad f(-1) = 2, \quad f(1) = -2, \quad f(2) = 2.$$

Hence

$$\arg \max_{x \in S} f(x) = \{-1, 2\}, \quad \max_{x \in S} f(x) = 2.$$

The point -1 is a local and global maximum; 2 is a boundary local maximum and a global maximum. The point 1 is a local and global minimum. The $\arg\max$ records the choices attaining the value; the maximum records the attained value itself.

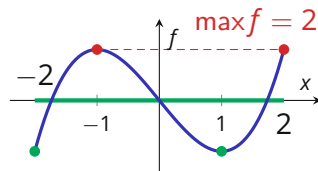


Figure 5 · $f(x) = x^3 - 3x$ on $[-2, 2]$: two maximizers, one maximum value.

Section 2

Existence, FOC & SOC

Does an optimization problem attain its optimum?

Here, existence means that at least one feasible choice attains the largest objective value:

$$\arg \max_{x \in S} f(x) \neq \emptyset.$$

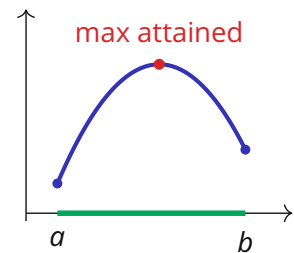
- ◆ Not every optimization problem has a maximizer: $\max_{x \in \mathbb{R}} x$ has none.
- ◆ From Functions: if $S \subseteq \mathbb{R}^n$ is nonempty and compact and $f : S \rightarrow \mathbb{R}$ is continuous, the Extreme Value Theorem (Weierstrass) guarantees a maximizer $x^* \in S$.
- ◆ Concavity serves a different role: it helps turn local conditions into global conclusions.

Logical order: first ask whether a solution exists; then derive optimality conditions and distinguish necessity from sufficiency; then use curvature, or compare the candidate values directly, to establish global optimality and uniqueness.

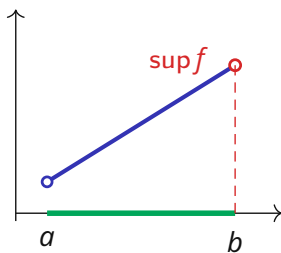
Further reading: Boyd, §29.42+ and §30.2; Carter, Ch. 6, pp. 601–602; §6.2, pp. 609–621.

Video: Mike, the Mathematician, *Continuous Functions on Compact Sets and the Extreme Value Theorem*.

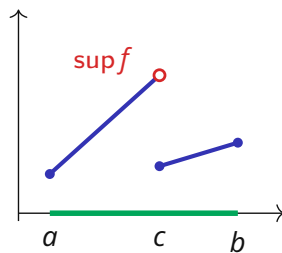
Attained maximum versus unattained supremum



closed interval,
continuous f : attained



open interval:
supremum at
the open end



jump at c :
supremum at
the missing limit

Figure 6 · Left: a compact feasible set and a continuous objective attain the maximum. Middle: an open endpoint removes the point where the supremum would be reached. Right: a jump does the same on a closed interval. Each hypothesis of the Extreme Value Theorem blocks one failure.

Consumer choice: why a maximizer exists

A standard linear budget set with strictly positive prices is compact

Let $p \gg 0$, $w > 0$, and $B(p, w) = \{x \in \mathbb{R}_+^n : p \cdot x \leq w\}$.

Closed. The map $x \mapsto p \cdot x$ is continuous, so from Functions $\{x \in \mathbb{R}^n : p \cdot x \leq w\}$, its preimage of the closed set $(-\infty, w]$, is closed; so is each $\{x : x_i \geq 0\}$. Their intersection $B(p, w)$ is closed, since its complement is a union of open sets.

Bounded. If $x \in B(p, w)$, then $0 \leq x_i \leq w/p_i$ for each i .

Compact. Closed and bounded in $\mathbb{R}^n$ is compact (Heine–Borel, Functions), so for every continuous $u : B(p, w) \rightarrow \mathbb{R}$ the Extreme Value Theorem gives $\arg \max_{x \in B(p, w)} u(x) \neq \emptyset$.

Linear Algebra supplies the inner product $p \cdot x$; Functions supplies the preimage, closed and bounded sets, compactness and the EVT; the argmax is the set defined on the first Definitions page.

Compactness versus convexity

Compactness can fail.

With $p \gg 0$, $w > 0$, the set $\{x \in \mathbb{R}_+^n : p \cdot x < w\}$ is bounded and misses the limit of $(1 - 1/k)(w/p_1)e_1$, so it is not closed.

At $p = (1, 0)$, $w = 1$, the set $\{x \in \mathbb{R}_+^2 : x_1 \leq 1\}$ is closed and unbounded.

Convexity can fail.

$\{x \in [0, 1]^2 : x_1 x_2 = 0\}$, the two axis segments, is compact, but the midpoint of $(1, 0)$ and $(0, 1)$ leaves it.

Linear inequalities always give convexity, and $x_1^2 + x_2^2 \leq 1$ shows a nonlinear constraint can too.

With $p \gg 0$, the weak inequality makes $B(p, w)$ closed, positive prices make it bounded, and linearity makes it convex. Existence uses compactness with continuity; convexity is used later, for globality and uniqueness.

Stationarity

FONC. If f is **differentiable** at x^* and x^* is an **interior** local extremum, then

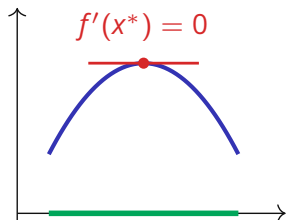
$$\nabla f(x^*) = 0.$$

1. Solve $\nabla f(x) = 0$ to find interior critical points.
2. Check the **boundary** and **non-differentiable** points separately.
3. Compare objective values when several feasible possibilities remain.

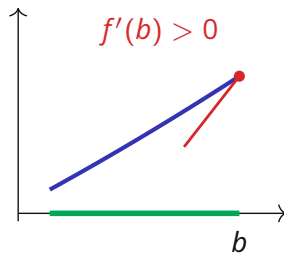
Reduce to one variable: fix i and set $\varphi_i(t) = f(x^* + te_i)$, with e_i the i th coordinate vector. Since x^* is interior, φ_i is defined on an interval around 0 and has a local extremum at $t = 0$, so $\varphi_i'(0) = 0$. The chain rule from Differentiation gives $\varphi_i'(0) = \nabla f(x^*)^\top e_i = \partial f(x^*)/\partial x_i$. Every partial derivative vanishes, so $\nabla f(x^*) = 0$. Stationarity is necessary; existence and optimality require separate checks.

Video: [Krista King, Critical points of multivariable functions.](#)

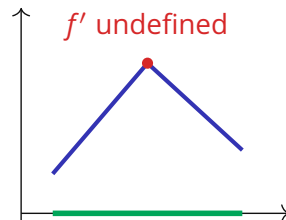
Three places a maximum can sit



interior: stationary



boundary: check separately



kink: check separately

Figure 7 · Interior maximum with a horizontal tangent; boundary maximum where the derivative is positive; kink where the derivative fails to exist. Solving $f'(x) = 0$ finds the first case only, which is why the boundary and the kinks are checked separately.

At an interior critical point

For f twice continuously differentiable near x^* , the second-order Taylor expansion at $x^* + h$ is

$$f(x^* + h) - f(x^*) = \nabla f(x^*)^\top h + \frac{1}{2} h^\top H_f(x^*) h + o(\|h\|^2).$$

At an interior critical point, $\nabla f(x^*) = 0$, so

$$f(x^* + h) - f(x^*) = \frac{1}{2} Q(h) + o(\|h\|^2), \quad Q(h) := h^\top H_f(x^*) h.$$

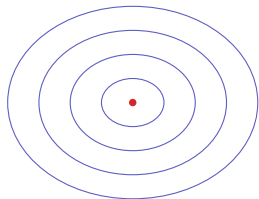
Local maximum. Necessary: $H(x^*) \preceq 0$.
Sufficient: $H(x^*) \prec 0$.

Local minimum. Necessary: $H(x^*) \succeq 0$.
Sufficient: $H(x^*) \succ 0$.

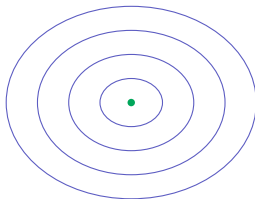
This combines Taylor expansion and the Hessian from Differentiation with the quadratic-form sign test from Convexity. Indefinite gives a saddle; **semidefinite can leave the test inconclusive**.

Video: [Dr. Trefor Bazett, Multi-variable optimization and the second derivative test.](#)

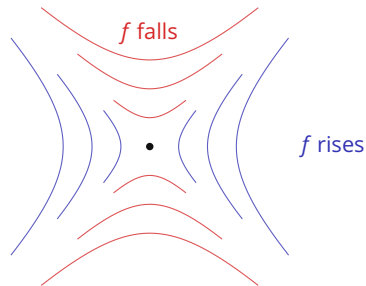
Level curves near a critical point



$H \prec 0$: local max
 f falls outward



$H \succ 0$: local min
 f rises outward



H indefinite: saddle
 f rises along one axis, falls along the other

Figure 8 · Level curves of f around a critical point. Negative definite Hessian: closed curves with f falling outward, a local maximum. Positive definite: f rising outward, a local minimum. Indefinite: hyperbolas, f rises along one axis and falls along the other, a saddle.

Local test versus global test

Suppose $S \subseteq \mathbb{R}^n$ is convex and f is twice continuously differentiable on an open set containing S . The same Hessian appears in both tests; the quantifier is what changes.

Convexity on S

$H_f(x) \succeq 0$ **for every** $x \in S \Rightarrow f$ is convex on S .

If $H_f(x) \succ 0$ **for every** $x \in S$, then f is strictly convex on S .

Concavity on S

$H_f(x) \preceq 0$ **for every** $x \in S \Rightarrow f$ is concave on S .

If $H_f(x) \prec 0$ **for every** $x \in S$, then f is strictly concave on S .

Convexity applied. The Hessian criterion from Convexity is a statement over the whole feasible set: once its sign establishes convexity or concavity on all of S , the local-to-global and uniqueness conclusions from “Local optima are global” apply.

Definiteness of $H_f(x^*)$ at the single point x^* gives only a local conclusion, and a boundary optimum need not satisfy $\nabla f(x^*) = 0$.

Further reading: Video: [Steve Brunton, Convexity 101](#). Boyd, Ch. 21.

One point test, one global test

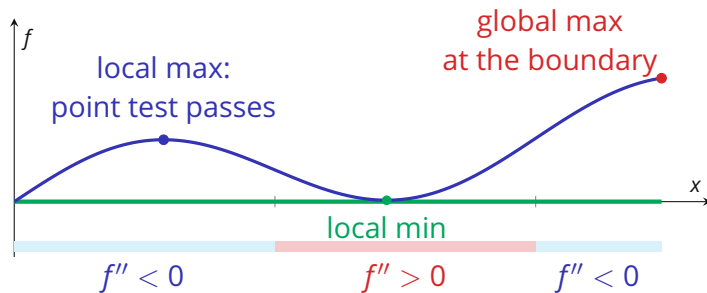


Figure 9 · At the local maximum $f'' < 0$, so the point test passes. The sign of f'' changes on S , so f is concave on only part of S and the local conclusion stays local: the global maximum sits at the boundary.

A quadratic objective

On $(x_1, x_2) \in [0, 4]^2$, let

$$f(x_1, x_2) = 2x_1x_2 - x_1^2 - 2x_2^2 + 3x_1 - 2x_2.$$

First-order conditions

$$\nabla f(x) = 0 \quad \Rightarrow \quad (x_1^*, x_2^*) = (2, \tfrac{1}{2}).$$

Second-order test

$$H_f = \begin{pmatrix} -2 & 2 \\ 2 & -4 \end{pmatrix}.$$

$$D_1 = -2 < 0 \text{ and } D_2 = 4 > 0, \text{ so } H_f \prec 0.$$

The constant Hessian is negative definite, so f is strictly concave on $[0, 4]^2$. Hence $(2, \frac{1}{2})$ is the unique global maximizer. Since f is quadratic, its second-order Taylor formula has no higher-order remainder.

Level curves of the quadratic objective

$$f(x_1, x_2) = 2x_1x_2 - x_1^2 - 2x_2^2 + 3x_1 - 2x_2$$

on $[0, 4]^2$.

- ◆ The level curves are ellipses centered at the maximizer $(2, \frac{1}{2})$.
- ◆ Their common shape is set by the constant Hessian: a negative definite H_f gives closed curves with f falling outward.

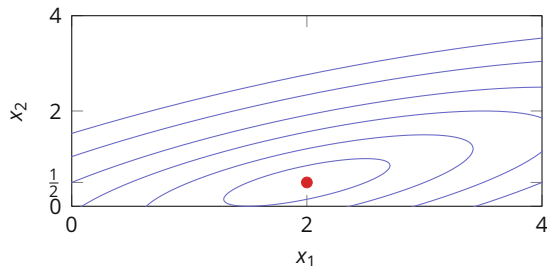


Figure 10 · Level curves of f on $[0, 4]^2$ and the maximizer $(2, \frac{1}{2})$.

Linear Algebra applied: FOC as a linear system

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 5 \\ 4 \end{bmatrix}, \quad f(x) = b^\top x - \frac{1}{2} x^\top A x.$$

- (a) Show that the FOC is the linear system $Ax = b$.
- (b) Compute A^{-1} and the critical point.
- (c) Use the Hessian to determine whether the critical point is the unique global maximizer.

Linear Algebra applied: FOC as a linear system

Since A is symmetric,

$$\nabla f(x) = b - Ax, \quad \nabla f(x) = 0 \Rightarrow Ax = b.$$

Moreover,

$$\det A = 3, \quad A^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \quad x^* = A^{-1}b = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

The Hessian is $-A$. The leading principal minors of A are $2 > 0$ and $3 > 0$, so $A \succ 0$, $-A \prec 0$, and f is strictly concave. Thus x^* is the unique global maximizer.

Necessary and sufficient conditions

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be twice differentiable and let x^* be an interior point. Use the implication definitions of necessary and sufficient conditions from Preliminaries. For the statement " x^* is a strict local maximum," classify each condition as necessary, sufficient, both, or neither:

$$(a) f'(x^*) = 0, \quad (b) f''(x^*) < 0,$$

$$(c) f'(x^*) = 0 \text{ and } f''(x^*) < 0.$$

Give a counterexample for every failed implication.

Necessary and sufficient conditions

- (a) Necessary but not sufficient. Fermat's theorem gives necessity; $f(x) = x^3$ at $x^* = 0$ shows that stationarity is not sufficient.
- (b) Neither. The function $f(x) = -x^4$ has a strict maximum at zero with $f''(0) = 0$. Conversely, $f(x) = x - x^2$ has $f''(0) < 0$, but zero is not a local maximum because $f'(0) = 1$.
- (c) Sufficient but not necessary. The usual derivative test using f'' gives sufficiency, while $f(x) = -x^4$ again shows failure of necessity.

Boundary points, saddles, and an inconclusive SOC

- (a) Maximize $f(x) = -(x - 3)^2$ on $[0, 2]$. Identify the role of the boundary.
- (b) Classify the critical point of $g(x, y) = x^2 - y^2$.
- (c) At $x = 0$, the second derivative of both $h(x) = x^4$ and $k(x) = -x^4$ is zero. Classify the point for each function and state what the SOC concludes.

Boundary points, saddles, and an inconclusive SOC

On $[0, 2]$, $f'(x) = 6 - 2x > 0$, so f is increasing and its maximum is $f(2) = -1$ at the boundary $x = 2$; there is no interior stationary point. For g , the only critical point is $(0, 0)$, and

$$H_g = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$$

is indefinite, so $(0, 0)$ is a saddle. Finally, $h(x) \geq 0 = h(0)$, so 0 is a strict global minimum of h ; $k(x) \leq 0 = k(0)$, so 0 is a strict global maximum of k . In both cases the zero second derivative makes the SOC inconclusive.

Section 3

Equality constraints

The Lagrangian

Problem. $\max_{x \in \mathbb{R}^n} f(x)$ subject to $g_j(x) = b_j, j = 1, \dots, m$.

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{j=1}^m \lambda_j (b_j - g_j(x)).$$

- ◆ If the equality-constraint gradients are **linearly independent at a constrained local optimum**, equivalently the constraint Jacobian has full row rank, then $\nabla_x \mathcal{L} = 0$, together with $g_j(x) = b_j$.
- ◆ Substitute the constraint directly when this is simpler.
- ◆ If f is concave and the equality constraints are affine, the FOCs are sufficient for a global maximum.

Further reading: Boyd, Ch. 18.

Video: Dr. Trefor Bazett, *Lagrange Multipliers: Geometric Meaning and Example*.

A first Lagrange problem, solved in full

Problem and Lagrangian

$$\max_{x,y} 6x + 4y - x^2 - y^2 \quad \text{s.t.} \quad x + y = 4,$$

$$\mathcal{L} = 6x + 4y - x^2 - y^2 + \lambda(4 - x - y).$$

First-order conditions

$$6 - 2x - \lambda = 0, \quad 4 - 2y - \lambda = 0, \quad x + y = 4.$$

These are linear in (x, y, λ) :

$$\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ \lambda \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 4 \end{bmatrix}.$$

Solve

Eliminate as in Linear Algebra: subtracting the second equation from the first removes λ :

$2 - 2x + 2y = 0$, so $x - y = 1$. With $x + y = 4$,

$$x^* = \frac{5}{2}, \quad y^* = \frac{3}{2}, \quad \lambda^* = 6 - 2x^* = 1.$$

Why this is the maximum

$H_f = \text{diag}(-2, -2) \prec 0$: f is strictly concave and the constraint is affine, so the FOCs are sufficient (previous page) and (x^*, y^*) is the unique global maximizer, with value $\frac{25}{2}$.

Four steps, repeated in every Lagrange problem: write $\mathcal{L}$, differentiate, solve together with the constraint, justify optimality.

When the Lagrange condition can fail

A maximizer with no multiplier

Consider

$$\max_x x \quad \text{s.t.} \quad x^2 = 0.$$

The feasible set is $\{0\}$, so $x^* = 0$ is the unique global maximizer. However,

$$f'(0) = 1, \quad g'(0) = 0,$$

so no multiplier λ satisfies $f'(0) = \lambda g'(0)$. The constraint gradient vanishes at the optimum, and the Lagrange multiplier condition fails. The linear independence of the constraint gradients required on the previous page rules this case out.

Tangency of level curve and constraint

At an interior constrained optimum with $\nabla g(x^*) \neq 0$ and $\nabla f(x^*) \neq 0$, the objective level curve and the constraint curve are tangent:

$$\nabla f(x^*) = \lambda^* \nabla g(x^*).$$

- ◆ The constraint is the level set $g^{-1}(\{b\})$ from Functions, and from Differentiation a gradient is normal to its level set. Tangent level sets have collinear normals.
- ◆ This condition pins down λ^* ; its interpretation comes later in this section.

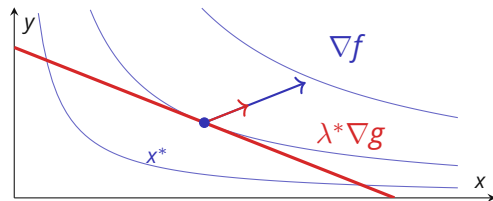


Figure 11 · Level curves of f (blue), the constraint $g = b$ (red), and the collinear gradients at x^* .

The level curve through x^* and the constraint are tangent there.

Second-order conditions along feasible directions

Feasible directions and restricted curvature

Consider

$$\max_x f(x) \quad \text{s.t.} \quad h(x) = 0,$$

where f and $h : \mathbb{R}^n \rightarrow \mathbb{R}^m$ are twice continuously differentiable near x^* . Let (x^*, λ^*) satisfy the FOCs and suppose $Dh(x^*)$ has full row rank. Write

$$H_{\mathcal{L}} = \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*).$$

A first-order feasible direction d is tangent to the constraint set when

$$Dh(x^*)d = 0.$$

Second-order conditions along feasible directions

Curvature on the tangent space

Necessary for max $d^\top H_{\mathcal{L}} d \leq 0$ for every tangent direction d .

Sufficient for max $d^\top H_{\mathcal{L}} d < 0$ for every nonzero tangent direction d .

Unconstrained SOC test curvature in every direction; equality-constrained SOC test directions that preserve the constraints to first order. The bordered Hessian is a computational device for this test and is left to the formal Optimization course.

Further reading: Boyd, §19.2; MIT 6.7220, Lecture 7.

Linear Algebra applied: feasible directions

Kernel and normal space

For affine constraints $Ax = b$, a displacement d remains feasible precisely when

$$A(x + d) = b \quad \Leftrightarrow \quad Ad = 0.$$

Hence feasible directions form $\ker(A)$. With $\mathcal{L}(x, \lambda) = f(x) + \lambda^\top (b - Ax)$, stationarity gives

$$\nabla f(x^*) = A^\top \lambda^*.$$

Therefore, for every feasible direction d ,

$$\nabla f(x^*)^\top d = (\lambda^*)^\top Ad = 0.$$

Thus $\nabla f(x^*)$ is orthogonal to every feasible direction $d \in \ker(A)$.

Feasible directions and restricted curvature

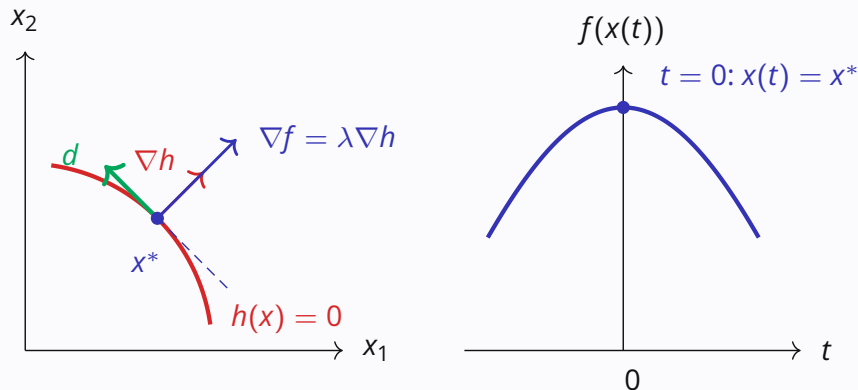


Figure 12 · Left: at x^* the tangent direction d satisfies $Dh(x^*)d = 0$, ∇h and ∇f are collinear, and the dashed level curve of f is tangent to the constraint. Right: along the constraint, $t \mapsto f(x(t))$ peaks at $t = 0$; $d^\top H_{\mathcal{L}} d < 0$ is its curvature there.

Quadratic Lagrange equations

A block linear system

Consider

$$\min_x \frac{1}{2}x^\top Qx - c^\top x \quad \text{s.t.} \quad Ax = b,$$

with $Q = Q^\top$, and use $\mathcal{L} = \frac{1}{2}x^\top Qx - c^\top x + \lambda^\top (Ax - b)$. The stationarity and feasibility equations are

$$Qx - c + A^\top \lambda = 0, \quad Ax = b.$$

They form the linear system

$$\begin{bmatrix} Q & A^\top \\ A & 0 \end{bmatrix} \begin{bmatrix} x \\ \lambda \end{bmatrix} = \begin{bmatrix} c \\ b \end{bmatrix}.$$

Output on a technology frontier

A firm chooses positive inputs (x, y) and production is $Q(x, y) = x^2 y$. The feasible frontier is

$$2x^2 + y^2 = 3.$$

The Lagrangian is

$$\mathcal{L} = x^2 y + \lambda(3 - 2x^2 - y^2),$$

with first-order conditions

$$2xy - 4\lambda x = 0, \quad x^2 - 2\lambda y = 0, \quad 2x^2 + y^2 = 3.$$

With $x > 0$ the first equation gives $\lambda = y/2$, the second then gives $x^2 = y^2$, so $x = y$, and the frontier gives $x^* = y^* = 1$, $\lambda^* = \frac{1}{2}$. Zero input gives zero output on the boundary, so $(1, 1)$ is the maximizer.

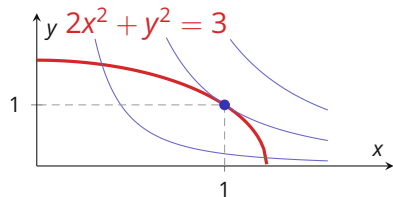


Figure 13 · Isoquants $x^2 y = c$ (blue) and the frontier (red), tangent at $(1, 1)$.

Shadow prices

With the sign convention $\mathcal{L} = f + \sum_j \lambda_j(b_j - g_j)$, if the value function V is differentiable, the envelope theorem gives

$$\lambda_j^* = \frac{\partial V}{\partial b_j}.$$

Thus the multiplier is the local marginal value of increasing the corresponding right-hand side b_j ; when larger b_j means more resources, this is the value of relaxing the constraint.

- ◆ Consumer theory: λ is the marginal utility of income.
- ◆ Firm theory: λ is the marginal value of relaxing a resource constraint.

The Envelope theorem section derives this identity.

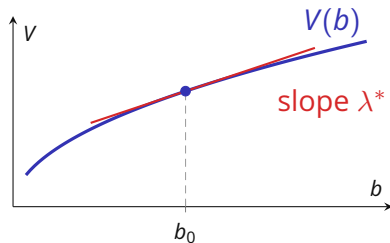


Figure 14 · The value function $V(b)$ and its tangent at b_0 :
the slope is the multiplier λ^* .

Linear Algebra and Differentiation: least squares

From squared error to normal equations

Begin with the unrestricted problem

$$\min_{\beta \in \mathbb{R}^k} S(\beta), \quad S(\beta) = (y - X\beta)^\top (y - X\beta).$$

Expanding the objective gives

$$S(\beta) = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X \beta.$$

Since $X^\top X$ is symmetric,

$$\nabla_\beta S(\beta) = -2X^\top y + 2X^\top X \beta.$$

The FOC is the system of normal equations

$$X^\top X \hat{\beta} = X^\top y.$$

Linear Algebra and Convexity: least squares

Solving the normal equations

If X has full column rank, then $X^\top X$ is positive definite and invertible. Hence

$$\hat{\beta} = (X^\top X)^{-1} X^\top y.$$

The Hessian is

$$H_S = 2X^\top X \succ 0,$$

so the critical point is the unique global minimizer. Full column rank and invertibility come from Linear Algebra; positive definiteness converts the Hessian calculation into strict convexity. Restricted least squares adds the affine condition $R\beta = r$; its Lagrange equations combine the normal equations with that restriction.

Least squares as a projection

The normal equations $X^\top (y - X\hat{\beta}) = 0$ say that the residual is orthogonal to every column of X .

Hence $X\hat{\beta}$ is the point of $\text{col}(X)$ nearest to y : minimizing $\|y - X\beta\|^2$ is projecting y onto the column space.

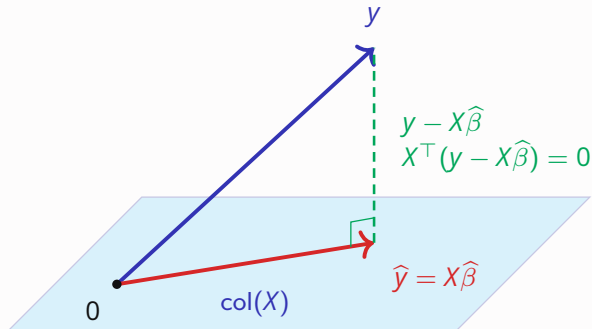


Figure 15 · The fitted vector is the orthogonal projection of y onto $\text{col}(X)$.

Restricted least squares

Constrained least-squares

Econometrics often imposes linear equality restrictions on coefficients:

$$y = X\beta + u, \quad R\beta = r.$$

The constrained least-squares problem is

$$\min_{\beta} (y - X\beta)'(y - X\beta) \quad \text{s.t.} \quad R\beta = r.$$

Examples include equal treatment effects, zero long-run restrictions, and adding-up restrictions. For instance, with two coefficients, $\beta_1 = \beta_2$ is $R = \begin{bmatrix} 1 & -1 \end{bmatrix}$ and $r = 0$.

Likelihood ratio test

Restricted likelihood

A likelihood-ratio test compares the best unrestricted fit with the best fit under restrictions:

$$LR = 2(\ell(\hat{\theta}) - \ell(\hat{\theta}_R)).$$

Unrestricted. Maximize $\ell(\theta)$ over all admissible θ .

Restricted. Maximize $\ell(\theta)$ subject to $R\theta = r$.

The statistic measures how much optimized value is lost when the equality restrictions are imposed. For instance, $\ell(\hat{\theta}) = -100$ and $\ell(\hat{\theta}_R) = -103$ give $LR = 2[-100 - (-103)] = 6$.

Lagrange multipliers

Solve both problems with a Lagrangian: write $\mathcal{L}$, solve the first-order conditions together with the constraint, report (x^*, y^*, λ^*) , and say why the stationary point is the global optimum.

(a) $\max_{x,y>0} xy \quad \text{s.t.} \quad x + y = 10.$

(b) $\min_{x,y \in \mathbb{R}} x^2 + y^2 \quad \text{s.t.} \quad x + 2y = 4.$

The product problem

For (a), use $\mathcal{L} = xy + \lambda(10 - x - y)$. The FOCs are $y - \lambda = 0$ and $x - \lambda = 0$, so $x = y = \lambda$; the constraint then gives

$$(x^*, y^*, \lambda^*) = (5, 5, 5).$$

The Hessian of xy is $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, indefinite, so concavity is unavailable as the reason for optimality. Substitute the constraint instead: on the feasible set $y = 10 - x$ with $0 < x < 10$, the objective is $\varphi(x) = x(10 - x)$, with $\varphi'' = -2 < 0$, so $x^* = 5$ is its unique maximizer and $(5, 5)$ is the global constrained maximizer.

Distance to a line

For (b), use $\mathcal{L} = x^2 + y^2 + \lambda(4 - x - 2y)$. The FOCs are $2x - \lambda = 0$ and $2y - 2\lambda = 0$, so $x = \lambda/2$ and $y = \lambda$; the constraint gives $\lambda/2 + 2\lambda = 4$. Hence

$$(x^*, y^*, \lambda^*) = \left(\frac{4}{5}, \frac{8}{5}, \frac{8}{5} \right).$$

The objective is strictly convex and the constraint is affine, so this stationary point is the unique global minimizer (the minimization mirror of the concave case on the Lagrangian page). Geometrically, $(x^*, y^*) = \frac{4}{5}(1, 2)$ is parallel to the constraint normal: the point of the line nearest the origin lies on the normal through the origin.

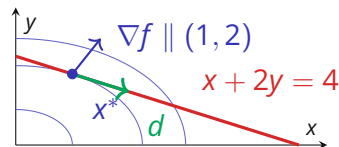


Figure 16 · Level circles of $x^2 + y^2$, the constraint line, the normal direction $(1, 2)$, and the direction $(2, -1)$ along the line.

Feasible directions and the block system

Second-order checks for the two Lagrange problems

For the two problems of the preceding exercise:

- (a) Find a nonzero feasible direction d with $Dh(x^*)d = 0$ and evaluate $d^\top H_{\mathcal{L}}(x^*, \lambda^*)d$.
What does the sign confirm in each case?
- (b) Write the FOCs of the distance problem as a block linear system in (x, y, λ) .

Feasible directions and the block system

Restricted curvature has the right sign in both problems

Product problem. $H_{\mathcal{L}} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and feasible directions satisfy $d_1 + d_2 = 0$. With $d = (t, -t)$,

$$d^T H_{\mathcal{L}} d = -2t^2 < 0 \quad (t \neq 0),$$

confirming a strict constrained maximum although $H_{\mathcal{L}}$ itself is indefinite.

Distance problem. Feasible directions form $\ker \begin{bmatrix} 1 & 2 \end{bmatrix} = \text{span}\{(2, -1)\}$, the direction drawn in Figure 16. With $d = (2t, -t)$ and $H_{\mathcal{L}} = 2I$, $d^T H_{\mathcal{L}} d = 10t^2 > 0$: a strict constrained minimum. Stationarity and the constraint form

$$\begin{bmatrix} 2 & 0 & -1 \\ 0 & 2 & -2 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ \lambda \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}.$$

This is the block system of the quadratic Lagrange page with $Q = 2I$, $A = \begin{bmatrix} 1 & 2 \end{bmatrix}$, $c = 0$, and $b = 4$. The multiplier column has the opposite sign because our Lagrangian uses $\lambda(b - Ax)$.

Section 4

Inequality constraints (KKT)

Binding or slack

Problem. $\max_{x \in \mathbb{R}^n} f(x)$ subject to $g_j(x) \leq b_j$.

At an optimum each constraint is either

Binding $g_j(x) = b_j$, or

Slack $g_j(x) < b_j$.

Resource constraint $x + y \leq 10$: plan (4, 6) binds;
plan (3, 5) leaves two units unused and is slack.

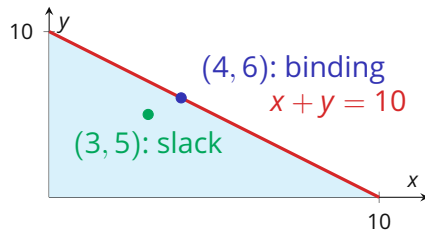


Figure 17 · The constraint $x + y \leq 10$: (4, 6) lies on the boundary, (3, 5) inside.

KKT conditions

For $\max_x f(x)$ subject to $g_j(x) \leq b_j$, use $\mathcal{L}(x, \lambda) = f(x) + \sum_j \lambda_j(b_j - g_j(x))$. At a local optimum where a constraint qualification holds, there are multipliers $\lambda^* \geq 0$ satisfying

Four conditions.

Stationarity	$\nabla_x \mathcal{L}(x^*, \lambda^*) = 0,$
Feasibility	$g_j(x^*) \leq b_j,$
Dual feasibility	$\lambda_j^* \geq 0,$
Complementary slackness	$\lambda_j^*(b_j - g_j(x^*)) = 0.$

Video: aa4cc, *Inequality-constrained optimization: KKT conditions*.

Normals of binding linear constraints

A geometric reading of stationarity

Suppose the constraints are $a_j^\top x \leq b_j$. KKT stationarity becomes

$$\nabla f(x^*) = \sum_{j=1}^m \lambda_j^* a_j, \quad \lambda_j^* \geq 0.$$

Complementary slackness sets $\lambda_j^* = 0$ whenever constraint j is slack. Thus only normals of binding constraints can contribute to the objective gradient. At a corner, several normals can be needed; at a smooth boundary point, one normal may suffice.

If a feasible direction d points into every binding constraint, then $a_j^\top d \leq 0$. Hence $\nabla f(x^*)^\top d \leq 0$: moving a small distance along d cannot raise the objective to first order.

Objective gradient in the cone of binding normals

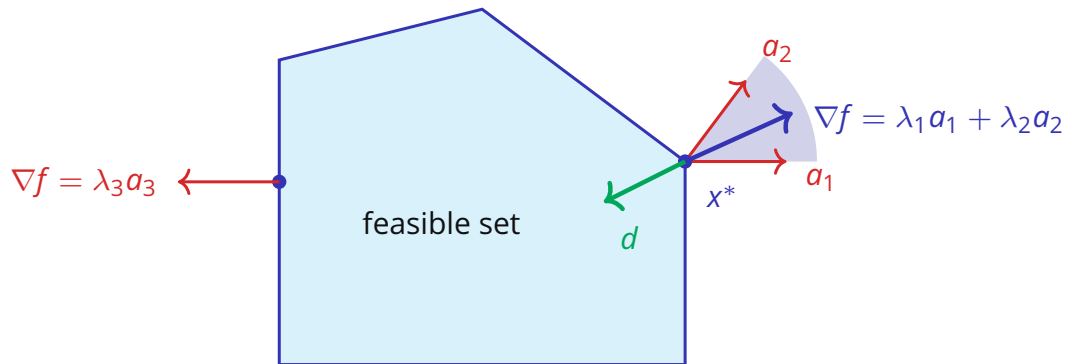


Figure 18 · At the corner x^* two constraints bind and ∇f lies in the shaded cone spanned by their outward normals a_1, a_2 ; at a smooth boundary point one normal suffices. A feasible direction d points into the set and makes an obtuse angle with ∇f .

Complementary slackness

A product is zero if and only if one of its factors is zero, so $\lambda_j^* (b_j - g_j(x^*)) = 0$ holds if and only if $\lambda_j^* = 0 \vee g_j(x^*) = b_j$, with the inclusive “or” from Preliminaries: both can hold. Equivalently, $g_j(x^*) < b_j \Rightarrow \lambda_j^* = 0$ and $\lambda_j^* > 0 \Rightarrow g_j(x^*) = b_j$. The converses fail: a binding constraint may have a zero multiplier (Figure 19, right panel).

Where this sits in duality theory

Suppose strong duality holds and both optima are attained. The primal and dual solutions are then a saddle point of the Lagrangian: the primal variables maximize it, the multipliers minimize it. The primal and dual problems, the dual function and strong duality are developed in the formal Optimization course.

Further reading: Boyd, Ch. 18; Boyd–Vandenberghe, §5.4.

Videos: Arizona Math Camp 40(A), conceptual and geometric insight; 40(B), conditions and theorem.

Three ways complementary slackness can look

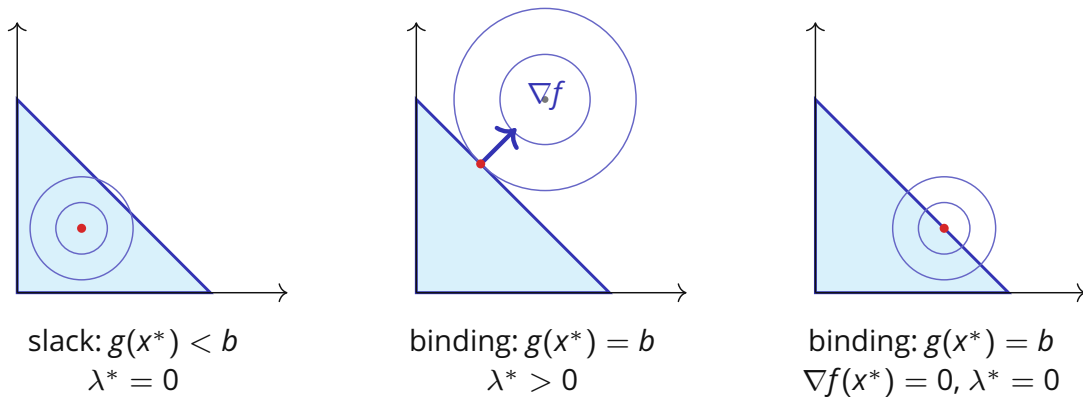


Figure 19 · Level circles of f over the same feasible triangle. Left: the peak is interior, the constraint is slack, the multiplier is zero. Middle: the peak lies outside, the optimum is pushed onto the constraint and the multiplier is positive. Right: the peak lies on the constraint line itself, so the constraint binds with a zero multiplier.

Nonnegativity: two equivalent KKT forms

Suppose the problem also requires $x_i \geq 0$. Write this as $-x_i \leq 0$ and let $\tilde{\mathcal{L}}(x, \lambda) = f(x) + \sum_j \lambda_j (b_j - g_j(x))$, omitting only the nonnegativity constraint.

With the multiplier μ_i .

$$\begin{aligned}\mathcal{L} &= \tilde{\mathcal{L}} + \mu_i x_i, \\ \frac{\partial \tilde{\mathcal{L}}}{\partial x_i} + \mu_i &= 0, \\ x_i \geq 0, \quad \mu_i &\geq 0, \quad \mu_i x_i = 0.\end{aligned}$$

Stationarity is an equality.

After eliminating μ_i . Since $\mu_i = -\partial \tilde{\mathcal{L}} / \partial x_i$,

$$\begin{aligned}x_i \geq 0, \quad \frac{\partial \tilde{\mathcal{L}}}{\partial x_i} &\leq 0, \\ x_i \frac{\partial \tilde{\mathcal{L}}}{\partial x_i} &= 0.\end{aligned}$$

If $x_i > 0$, the partial derivative is zero. If $x_i = 0$, it may be negative.

Further reading: [Columbia Math Camp, Constrained Optimization, Sec. 6.](#)

Minimization: the same system in the other sign convention

Standard minimization form

$$\begin{array}{ll} \min_x & f(x) \\ \text{s.t.} & g_i(x) \leq 0, \quad h_r(x) = 0. \end{array} \quad \mathcal{L} = f(x) + \sum_i \lambda_i g_i(x) + \sum_r \nu_r h_r(x), \quad \lambda_i \geq 0.$$

KKT system in the $g_i \leq 0$ form.

$$\begin{aligned} \nabla f(x^*) + \sum_i \lambda_i^* \nabla g_i(x^*) + \sum_r \nu_r^* \nabla h_r(x^*) &= 0, \\ g_i(x^*) \leq 0, \quad h_r(x^*) &= 0, \quad \lambda_i^* \geq 0, \quad \nu_r^* \in \mathbb{R}, \quad \lambda_i^* g_i(x^*) &= 0. \end{aligned}$$

Minimizing f is maximizing $-f$ on the same feasible set: this is the maximization system of the previous pages with the objective gradient and the multipliers relabelled, and $x_i \geq 0$ is handled by the same two equivalent forms.

Further reading: [Boyd–Vandenbergh](#); [MIT 6.7220](#).

Minimization: a solved KKT example

Problem and Lagrangian

$$\begin{aligned} \min_{x,y} \quad & (x-1)^2 + (y-1)^2 \\ \text{s.t.} \quad & x+y-3 \geq 0, \quad x, y \geq 0. \end{aligned}$$

Let $c(x,y) = x + y - 3$. Then

$$\mathcal{L} = (x-1)^2 + (y-1)^2 - \lambda c(x,y) - \mu_x x - \mu_y y.$$

If $c(x,y) > 0$, then $\lambda = 0$. The nonnegativity KKT system then gives $(x,y) = (1,1)$, which is infeasible. Thus $c(x,y) = 0$.

Further reading: [Tibshirani, CMU 10-725](#), *Karush-Kuhn-Tucker conditions*.

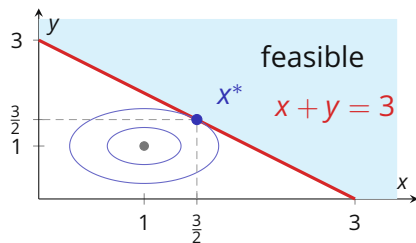


Figure 20 · Level circles around the target $(1,1)$, the feasible half-plane $x + y \geq 3$ (shaded), and the nearest feasible point x^* .

Minimization: KKT calculation

At the positive solution, $\mu_x = \mu_y = 0$. Stationarity and the binding constraint give

$$2(x - 1) - \lambda = 0, \quad 2(y - 1) - \lambda = 0, \quad x + y = 3.$$

The first two equations and $x + y = 3$ give

$$x = y = 1 + \frac{\lambda}{2}, \quad 3 = x + y = 2 + \lambda \Rightarrow \lambda = 1.$$

Substitution gives

$$(x^*, y^*) = \left(\frac{3}{2}, \frac{3}{2}\right), \quad \lambda^* = 1, \quad \mu_x^* = \mu_y^* = 0.$$

Primal and dual feasibility and complementary slackness hold. The objective is convex and the constraints define a convex feasible set, so this KKT point is the global minimizer.

Consumer choice: explicit multipliers

Consumer problem The consumer has perfect-substitutes utility:

$$\begin{aligned} \max_{x,y \geq 0} \quad & \alpha x + \beta y \\ \text{subject to} \quad & p_x x + p_y y \leq m, \end{aligned}$$

where $\alpha, \beta, p_x, p_y, m > 0$.

Slack-form Lagrangian

$$\mathcal{L} = \alpha x + \beta y + \lambda(m - p_x x - p_y y) + \mu_x x + \mu_y y.$$

Here λ is the marginal utility of income; μ_x and μ_y enforce nonnegativity.

KKT conditions

Stationarity

$$\alpha - \lambda p_x + \mu_x = 0, \quad \beta - \lambda p_y + \mu_y = 0.$$

Primal and dual feasibility

$$x, y \geq 0, \quad p_x x + p_y y \leq m, \quad \lambda, \mu_x, \mu_y \geq 0.$$

Complementary slackness

$$\lambda(m - p_x x - p_y y) = 0, \quad \mu_x x = \mu_y y = 0.$$

Consumer choice: reduced form

Eliminate each nonnegativity multiplier from stationarity and complementary slackness.

Good x

$$\alpha - \lambda p_x + \mu_x = 0,$$

$$\mu_x = \lambda p_x - \alpha \geq 0,$$

$$x(\alpha - \lambda p_x) = 0.$$

Good y

$$\beta - \lambda p_y + \mu_y = 0,$$

$$\mu_y = \lambda p_y - \beta \geq 0,$$

$$y(\beta - \lambda p_y) = 0.$$

Reduced KKT form.

$$x \geq 0, \quad \lambda p_x - \alpha \geq 0, \quad x(\lambda p_x - \alpha) = 0,$$

$$y \geq 0, \quad \lambda p_y - \beta \geq 0, \quad y(\lambda p_y - \beta) = 0,$$

$$m - p_x x - p_y y \geq 0, \quad \lambda \geq 0, \quad \lambda(m - p_x x - p_y y) = 0.$$

Consumer choice: corner solutions

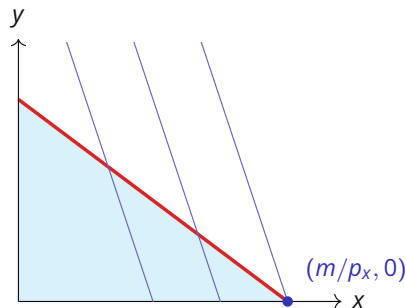
Since $\alpha, \beta > 0$, utility is increasing in both goods, so the budget binds: $p_x x + p_y y = m$.

A consumed good pins down the multiplier:

$x^* > 0 \Rightarrow \lambda^* = \alpha/p_x$ and $y^* > 0 \Rightarrow \lambda^* = \beta/p_y$.

At an x-corner, the unused good yields no more utility per dollar:

$$x^* > 0, y^* = 0 \Rightarrow \beta - \lambda^* p_y \leq 0 \Leftrightarrow \frac{\beta}{p_y} \leq \frac{\alpha}{p_x}.$$



$\alpha/p_x > \beta/p_y$: x-corner

Figure 21 · Perfect substitutes: indifference lines

$\alpha x + \beta y = k$ (blue) steeper than the budget line (red) put the optimum at the x-corner.

Video: [Economics in Many Lessons, Utility maximization: a corner solution.](#)

Consumer choice: compare the two goods

Compare marginal utility per dollar

$$\begin{aligned} \frac{\alpha}{p_x} &> \frac{\beta}{p_y} &\Rightarrow (x^*, y^*) &= \left(\frac{m}{p_x}, 0 \right), \\ \frac{\alpha}{p_x} &= \frac{\beta}{p_y} &\Rightarrow \{(x, y) \in \mathbb{R}_+^2 : p_x x + p_y y = m\}, \\ \frac{\alpha}{p_x} &< \frac{\beta}{p_y} &\Rightarrow (x^*, y^*) &= \left(0, \frac{m}{p_y} \right). \end{aligned}$$

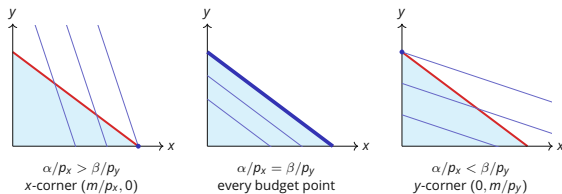


Figure 22 · The same comparison as slopes: indifference lines $\alpha x + \beta y = k$ (blue) against the budget line (red).

A KKT problem

Choose the feasible point closest to the target $(2, 3)$:

$$\max_{x, y \geq 0} 10 - [(x - 2)^2 + (y - 3)^2]$$

subject to $x + y \leq 3$ and $x + 2y \leq 6$.

- ◆ The unconstrained peak $(2, 3)$ is infeasible.
- ◆ Optimum: $(x^*, y^*) = (1, 2)$ and $(\lambda_1^*, \lambda_2^*) = (2, 0)$.
- ◆ Constraint 2 and both nonnegativity constraints are slack at the optimum.

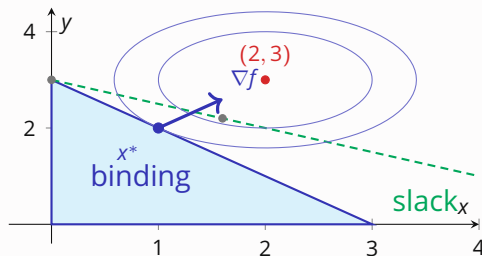


Figure 23 · Level circles of the objective, the feasible set, the slack constraint $x + 2y = 6$ (dashed), and $\nabla f = \lambda_1^*(1, 1)$ at x^* along the normal of the binding constraint $x + y = 3$.

The highest attainable level circle meets the feasible set at $x^* = (1, 2)$.

Solving the KKT conditions by cases

When $x, y > 0$, the nonnegativity multipliers are zero and stationarity is $-2(x - 2) - \lambda_1 - \lambda_2 = 0$ and $-2(y - 3) - \lambda_1 - 2\lambda_2 = 0$. If a case reaches $x = 0$ or $y = 0$, the corresponding nonnegativity multiplier must be restored.

No resource constraint binds

If $\lambda_1 = \lambda_2 = 0$, interior stationarity gives $(x, y) = (2, 3)$, which violates $x + y \leq 3$.

Both resource constraints bind

If $\lambda_1, \lambda_2 > 0$, both constraints bind, so $(x, y) = (0, 3)$. Since $y > 0$, its nonnegativity multiplier is zero; y -stationarity requires $-\lambda_1 - 2\lambda_2 = 0$, impossible for positive multipliers.

One resource constraint binds

First constraint

If $\lambda_1 > 0$ and $\lambda_2 = 0$, stationarity gives

$$x = 2 - \frac{\lambda_1}{2}, \quad y = 3 - \frac{\lambda_1}{2}.$$

Since $x + y = 3$,

$$5 - \lambda_1 = 3 \quad \Rightarrow \quad (x, y, \lambda_1) = (1, 2, 2),$$

and $x + 2y = 5 < 6$.

Second constraint

If $\lambda_1 = 0$ and $\lambda_2 > 0$, stationarity gives

$$x = 2 - \frac{\lambda_2}{2}, \quad y = 3 - \lambda_2.$$

Since $x + 2y = 6$,

$$8 - \frac{5}{2}\lambda_2 = 6 \quad \Rightarrow \quad (x, y, \lambda_2) = \left(\frac{8}{5}, \frac{11}{5}, \frac{4}{5}\right),$$

but $x + y = 19/5 > 3$.

Surviving KKT point: $(x^*, y^*, \lambda_1^*, \lambda_2^*) = (1, 2, 2, 0)$; here $x, y > 0$, so the nonnegativity multipliers are indeed zero.

When KKT conditions are necessary

Under differentiability and a **constraint qualification**, a local optimum satisfies KKT.

Further reading: Boyd–Vandenberghe, Ch. 5; MIT 6.7220, Lec. 7.

When KKT conditions are sufficient

If (x^*, λ) satisfies KKT, **f is concave and each g_j is convex**, and the sign convention is $\mathcal{L} = f + \sum_j \lambda_j(b_j - g_j)$, then x^* solves the problem globally.

Convexity applied. For convex g_j , each constraint set is the convex lower contour set $L_{g_j}(b_j) = \{x : g_j(x) \leq b_j\}$. Intersections of convex sets are convex, so the feasible set is convex. Concavity of f then turns the first-order inequalities into a global conclusion.

KKT sufficiency beyond concavity

Quasiconcave programming

For the convention $g_j(x) \leq b_j$, suppose f and the g_j are differentiable and each g_j is quasiconvex. If (x^*, λ^*) satisfies KKT, then x^* is a global maximizer when either

- (a) f is concave; or
- (b) f is quasiconcave and $\nabla f(x^*) \neq 0$.

The requirement $\nabla f(x^*) \neq 0$ in (b) is imposed at the KKT point. It excludes stationary points at which quasiconcavity gives no first-order ranking of feasible alternatives; the next page shows one.

Further reading: Walker, *Nonlinear Programming and the Kuhn–Tucker Conditions*.

Pseudoconcavity and first-order sufficiency

Pseudoconcavity

On a convex domain S , a differentiable $f : S \rightarrow \mathbb{R}$ is pseudoconcave when, for every $x, y \in S$,

$$f(y) > f(x) \Rightarrow \nabla f(x)^\top (y - x) > 0.$$

This implication directly converts the KKT gradient inequality for a feasible y into $f(y) \leq f(x^*)$. Hence a pseudoconcave objective and quasiconvex constraint functions also make KKT sufficient. Walker's KKT theorem needs $\nabla f(x^*) \neq 0$ only at the KKT point; his characterization of pseudoconcavity from quasiconcavity assumes $\nabla f(x) \neq 0$ throughout the domain.

Two comparisons. $f(x) = e^x$ is strictly convex, so it fails concavity, yet it is pseudoconcave on $\mathbb{R}$: $f(y) > f(x)$ forces $y > x$, and then $f'(x)(y - x) = e^x(y - x) > 0$. $f(x) = x^3$ is quasiconcave but fails pseudoconcavity at 0: $f(1) > f(0)$ while $f'(0)(1 - 0) = 0$. Its stationary point 0 satisfies KKT for $\max x^3$ and is beaten by every $x > 0$: the case that (b) on the previous page rules out.

Further reading: Walker, *Differentiable Quasiconcave Functions*.

Constraint qualifications

Constraint qualifications

Constraint qualifications rule out irregular local descriptions of the feasible boundary. In this brush-up, KKT necessity is invoked only when a constraint qualification holds.

In Optimization you will study constraint qualifications.

Further reading: Boyd, §19.4; MIT 6.7220, Lec. 7; Border, *Summary Notes on Maximization*, §6, pp. 11–12.

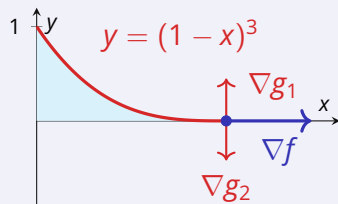


Figure 24 · $\max x$ s.t. $y \leq (1-x)^3$, $y \geq 0$: the optimum is the cusp $(1,0)$; $\nabla g_1 = (0,1)$ and $\nabla g_2 = (0,-1)$ are vertical, so their cone misses $\nabla f = (1,0)$.

Video: [Stanford EE364A, Lecture 8: Lagrangian duality and Slater's condition.](#)

Complementary slackness and shadow prices

Let

$$V(b) = \max_x f(x) \quad \text{s.t.} \quad g_j(x) \leq b_j,$$

and use $\mathcal{L} = f + \sum_j \lambda_j (b_j - g_j)$. Along a differentiable KKT solution path with a locally unchanged active set,

$$\frac{\partial V}{\partial b_j} = \lambda_j^*.$$

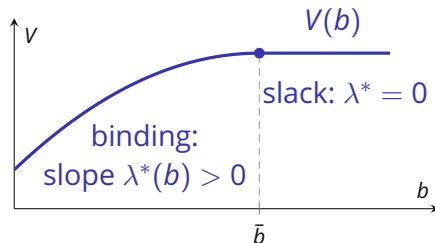


Figure 25 · One constraint, one right-hand side b : V rises with slope $\lambda^*(b) > 0$ while the constraint binds and is flat once it is slack; the active set changes at $\bar{b}$.

Which constraints have a positive shadow price

Complementary slackness ties each multiplier to the value of relaxing its constraint:

- ◆ If $g_j(x^*) < b_j$, then $\lambda_j^* = 0$, so $\partial V / \partial b_j = 0$: relaxing an unused constraint has zero first-order value.
- ◆ If $\lambda_j^* > 0$, then $g_j(x^*) = b_j$: the constraint binds and relaxing its right-hand side has positive marginal value.
- ◆ **Binding alone does not imply positive marginal value; a binding constraint may have $\lambda_j^* = 0$.**

Shadow prices in the two-resource example

In the two-resource example, $\lambda_1^* = 2$ and $\lambda_2^* = 0$: a marginal relaxation of the first resource constraint is worth 2, while the slack second constraint has zero shadow price.

The **parameter-envelope formula** for KKT problems is derived in the Envelope theorem section, after comparative statics separates movements in the optimizer from movements in optimized value.

KKT: a zero multiplier and a corner

Write the KKT conditions and solve each problem. Identify every binding constraint and multiplier.

(a) $\max_{x,y \geq 0} 4x + 2y - x^2 - y^2 \quad \text{s.t.} \quad x + y \leq 3.$

(b) $\max_{x,y \geq 0} 3x + y \quad \text{s.t.} \quad x + 2y \leq 6.$

KKT: a zero multiplier and a corner

Use $\mathcal{L} = f + \lambda(b - a^\top x) + \mu x + \nu y$, with nonnegative multipliers. In (a), strict concavity and stationarity give

$$(x^*, y^*) = (2, 1), \quad \lambda^* = \mu^* = \nu^* = 0.$$

The resource constraint binds because $2 + 1 = 3$, even though its multiplier is zero; both nonnegativity constraints are slack. In (b),

$$(x^*, y^*) = (6, 0), \quad \lambda^* = 3, \quad \mu^* = 0, \quad \nu^* = 5.$$

The resource constraint and $y \geq 0$ bind. The payoff per unit of the resource is larger for x , producing the corner solution.

KKT: necessity and sufficiency

Decide whether each statement is true or false and justify it.

- (a) At a differentiable local optimum, a constraint qualification makes KKT necessary.
- (b) For a concave maximization problem with convex inequality functions, any feasible KKT point is a global maximizer.
- (c) Any feasible point satisfying KKT is a global maximizer, regardless of curvature.

KKT: necessity and sufficiency

- (a) True. Differentiability plus a suitable constraint qualification gives the KKT necessary conditions at a local optimum.
- (b) True for the sign convention and hypotheses stated in this section: concavity of the objective and convexity of the functions defining the lower contour sets turn stationarity into $f(y) \leq f(x^*)$ for every feasible y .
- (c) False. For example, consider $\max_{-1 \leq x \leq 1} x^2$. At $x^* = 0$, both inequality constraints are slack, so their multipliers are zero, and stationarity holds because $f'(0) = 0$. Thus $x^* = 0$ satisfies KKT, but it is the global minimum; the global maximizers are $x = \pm 1$. Sufficiency of KKT needs the curvature assumptions of (b).

Section 5

Comparative statics

Parameterized problems: solution and value

For a parameter θ , let the feasible set be $S(\theta)$ and the objective be $f(x, \theta)$. Two distinct objects are

$$C(\theta) = \arg \max_{x \in S(\theta)} f(x, \theta),$$

$$V(\theta) = \sup_{x \in S(\theta)} f(x, \theta).$$

$C(\theta)$ is the **solution correspondence**: it may contain several maximizers.

$V(\theta)$ is the **value function**: when the maximum is attained, $V(\theta) = f(x^*, \theta)$ for every $x^* \in C(\theta)$.

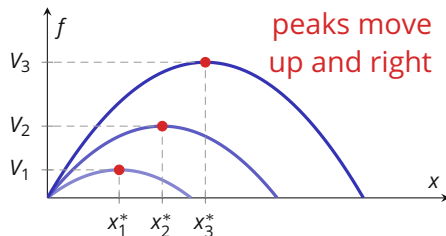


Figure 26 · Objectives $x \mapsto f(x, \theta)$ for $\theta_1 < \theta_2 < \theta_3$ (lighter to darker): the peaks' positions trace $C(\theta)$, their heights trace $V(\theta)$.

A solution correspondence can be set-valued

Let $x \in \{0, 1\}$ and $f(x, \theta) = x(\theta - k)$ with k fixed. Then

$$C(\theta) = \begin{cases} \{0\}, & \theta < k, \\ \{0, 1\}, & \theta = k, \\ \{1\}, & \theta > k, \end{cases} \quad V(\theta) = \max\{0, \theta - k\}.$$

The solution object is a correspondence: at $\theta = k$ the set $C(\theta)$ has two elements. A function $\theta \mapsto x^*(\theta)$ with $x^*(\theta) \in C(\theta)$ is a selection from C ; at $\theta = k$ the problem leaves that choice open, $x^*(k) = 0$ or $x^*(k) = 1$. Where $C(\theta)$ is a singleton the selection is forced. Every selection jumps from 0 to 1 as θ crosses k ; the value function is continuous, with a kink at k .

The next pages differentiate $x^*(\theta)$ and $V(\theta)$ along a differentiable solution path. This example has no such path at $\theta = k$: $C(k)$ has two elements and every selection jumps there.

Consumer demand and indirect utility

A consumer with prices $p \gg 0$ and wealth w has

$$B(p, w) = \{x \in \mathbb{R}_+^n : p \cdot x \leq w\},$$

$$C(p, w) = \arg \max_{x \in B(p, w)} u(x),$$

$$V(p, w) = \sup_{x \in B(p, w)} u(x).$$

If u is strictly quasiconcave on the convex budget set, $C(p, w)$ is a singleton whenever it is nonempty and can be written as an ordinary demand function. The optimizer set $C(p, w)$ and optimized value $V(p, w)$ are different objects.

Continuity properties of C and V belong to the later optimization course.

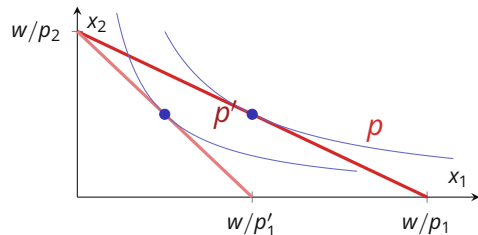


Figure 27 · A higher price p'_1 rotates the budget line inward; the optimum moves from the right dot, $C(p, w)$, to the left dot, $C(p', w)$, and the indifference level reached, V , falls.

Two sensitivity questions

For a parameterized problem, the solution object $C(\theta)$ and value $V(\theta)$ answer different questions.

Comparative statics.

How does the optimal choice change with the parameter? If the solution is unique and differentiable,

$$\theta \mapsto x^*(\theta), \quad \frac{dx^*}{d\theta}.$$

Using the implicit function theorem.

Envelope theorem.

How does the optimized value change with the parameter?

$$V(\theta) = f(x^*(\theta), \theta), \quad \frac{dV}{d\theta}.$$

If V is differentiable, the envelope theorem often avoids computing $dx^*/d\theta$.

Same parameterized problem, two different derivatives: one tracks the optimizer; the other tracks the optimized value.

Video: [Matt Clancy, Comparative statics](#).

Firm supply from an FOC

Interior output choice

A price-taking firm chooses an interior output $q^*(p)$ in $\max_{q>0} pq - c(q)$, with $c''(q) > 0$. The first-order condition $p - c'(q^*(p)) = 0$ holds at every price, so it is an identity in p .

Differentiate the identity

$$1 - c''(q^*(p)) \frac{dq^*}{dp} = 0, \quad \boxed{\frac{dq^*}{dp} = \frac{1}{c''(q^*(p))} > 0.}$$

This is the implicit function theorem from Differentiation applied to an optimization FOC: with strictly convex cost, a higher output price raises the firm's optimal output locally.

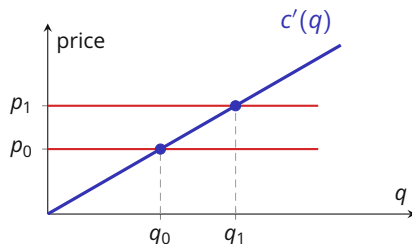


Figure 28 · Marginal cost $c'(q)$ and two price lines: the FOC $p = c'(q^*)$ is the crossing, a higher price moves it to the right, and the slope of c' at q^* is c'' .

Differentiate an FOC

Let $\theta > 0$ and consider

$$\max_{q>0} \left\{ \theta q - \frac{1}{3} q^3 \right\}.$$

- (a) Find $q^*(\theta)$ and differentiate it directly.
- (b) Differentiate the FOC implicitly, as for the firm's supply above, and verify the same derivative.

Differentiate an FOC

The FOC is $\theta - q^2 = 0$, and the objective is strictly concave on $q > 0$. Thus

$$q^*(\theta) = \sqrt{\theta}, \quad \frac{dq^*}{d\theta} = \frac{1}{2\sqrt{\theta}}.$$

Implicit differentiation of the identity $\theta - q^*(\theta)^2 = 0$ gives $1 - 2q^* \frac{dq^*}{d\theta} = 0$, hence

$$\frac{dq^*}{d\theta} = \frac{1}{2q^*(\theta)} = \frac{1}{2\sqrt{\theta}},$$

the same derivative, obtained without first solving for q^* .

Differentiation applied: local derivative of an optimizer

Differentiate the first-order condition

Suppose F is continuously differentiable near a point where an interior optimizer satisfies

$$F(x^*(\theta), \theta) = 0,$$

and $D_x F$ is invertible there. The implicit function theorem gives

$$\frac{dx^*}{d\theta} = -[D_x F(x^*, \theta)]^{-1} D_\theta F(x^*, \theta).$$

For $F = \nabla_x f$, continuous differentiability of F means f is twice continuously differentiable, and the matrix $D_x F$ is the Hessian with respect to x . The Jacobian comes from Differentiation; solving for $dx^*/d\theta$ uses matrix invertibility from Linear Algebra.

Further reading: [Boyd, Ch. 15; §19.3](#); [Border, Implicit Function Theorem](#).

Matrix comparative statics

Comparative statics in matrix form

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad b(\theta) = \begin{bmatrix} 5 + \theta \\ 4 - \theta \end{bmatrix},$$

and consider

$$\max_{x \in \mathbb{R}^2} \left\{ b(\theta)^\top x - \frac{1}{2} x^\top A x \right\}.$$

Find $x^*(\theta)$ and $dx^*/d\theta$. Then obtain the same derivative by differentiating the FOC $Ax^*(\theta) = b(\theta)$.

Matrix comparative statics

Two routes to the same derivative

Since $A^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$,

$$x^*(\theta) = A^{-1}b(\theta) = \begin{bmatrix} 2 + \theta \\ 1 - \theta \end{bmatrix}, \quad \frac{dx^*}{d\theta} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Differentiating the FOC gives

$$A \frac{dx^*}{d\theta} = \frac{db}{d\theta} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \frac{dx^*}{d\theta} = A^{-1} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

This is the formula of the preceding page with $F = b(\theta) - Ax$: $D_x F = -A$ and $D_\theta F = b'(\theta)$. Because $-A \prec 0$, this critical point is the unique global maximizer for every θ .

Increasing differences

Monotone comparative statics

For problems with corners or discrete choices, increasing differences ranks the gain from a higher x across values of θ :

$$x' > x, \quad \theta' > \theta \quad \Rightarrow \quad f(x', \theta') - f(x, \theta') \geq f(x', \theta) - f(x, \theta).$$

Higher θ raises the gain from higher x .

The general theory covers supermodularity, single crossing, lattices, and Topkis-type results.

In the binary example $f(x, \theta) = x(\theta - k)$, the gain from $x = 1$ over $x = 0$ is $f(1, \theta) - f(0, \theta) = \theta - k$, which rises with θ : increasing differences holds, and the optimal choice switches from 0 to 1 at $\theta = k$ with no FOC and no differentiable $x^*(\theta)$.

Further reading: [Echenique, Monotone Comparative Statics](#).

Increasing differences

The definition through a mixed partial derivative

- (a) Let $f(q, \theta)$ be twice continuously differentiable with $f_{q\theta} \geq 0$ everywhere. Show that f has increasing differences. *Hint:* integrate $f_{q\theta}$ over the rectangle $[q, q'] \times [\theta, \theta']$.
- (b) Verify increasing differences for $f(q, \theta) = \theta q - \frac{1}{3}q^3$ of the preceding exercise. Does $r(q, \theta) = -\theta q - \frac{1}{2}q^2$ have increasing differences?

Increasing differences

A nonnegative mixed partial derivative gives increasing differences

(a) For $q < q'$ and $\theta < \theta'$,

$$\int_{\theta}^{\theta'} \int_q^{q'} f_{q\theta}(s, t) ds dt = \int_{\theta}^{\theta'} [f_{\theta}(q', t) - f_{\theta}(q, t)] dt = [f(q', \theta') - f(q, \theta')] - [f(q', \theta) - f(q, \theta)].$$

The integrand is nonnegative, so the right side is nonnegative: the gain from raising q to q' is at least as large at θ' as at θ .

(b) For $\theta q - \frac{1}{3}q^3, f_{q\theta} = 1 > 0$, so increasing differences holds; this is consistent with $dq^*/d\theta > 0$ found in the preceding exercise. For $r, r_{q\theta} = -1$, and the double integral equals $-(q' - q)(\theta' - \theta) < 0$: higher θ lowers the gain from a higher q , so r lacks increasing differences.

Section 6

Envelope theorem

Differentiation applied: why the indirect effect disappears

Let an interior optimizer $x^*(\alpha)$, differentiable in α , define

$$V(\alpha) = \max_x f(x, \alpha) = f(x^*(\alpha), \alpha).$$

The chain rule gives

$$V'(\alpha) = f_x(x^*(\alpha), \alpha) \frac{dx^*}{d\alpha} + f_\alpha(x^*(\alpha), \alpha).$$

At an interior optimum, the FOC is

$$f_x(x^*(\alpha), \alpha) = 0.$$

Therefore

$$\begin{aligned} V'(\alpha) &= 0 \cdot \frac{dx^*}{d\alpha} + f_\alpha(x^*(\alpha), \alpha) \\ &= \boxed{f_\alpha(x^*(\alpha), \alpha)}. \end{aligned}$$

This is the chain rule from Differentiation applied at a point where the FOC makes the indirect first-order term zero.

Two derivatives from one parameterized problem

Comparative statics asks for $dx^*/d\alpha$. The envelope theorem asks for $dV/d\alpha$ and shows why that derivative can often be computed without solving for $dx^*/d\alpha$: the indirect first-order term vanishes at an interior optimum.

The name: for each fixed choice $\bar{x}$, the map $\alpha \mapsto f(\bar{x}, \alpha)$ is one curve. V is the upper envelope of these curves and touches, at each α , the curve of the current optimizer. That tangency is the vanishing indirect effect.

Further reading: [Boyd, §19.1](#). Video: [Arizona Math Camp, The Envelope Theorem and Shadow Values](#).

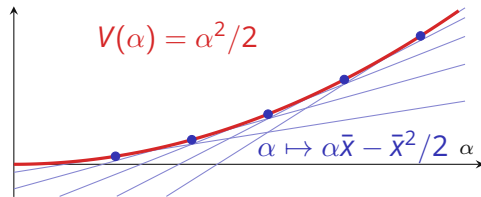


Figure 29 · $f(x, \alpha) = \alpha x - x^2/2$: for fixed $\bar{x}$ the map $\alpha \mapsto \alpha\bar{x} - \bar{x}^2/2$ is a line; $V(\alpha) = \alpha^2/2$ is their upper envelope, tangent to the line of $\bar{x} = \alpha$.

Equality-constrained derivation

Consider

$$V(\alpha) = \max_x f(x, \alpha) \quad \text{s.t.} \quad g(x, \alpha) = 0, \quad \mathcal{L} = f + \lambda(0 - g).$$

Along a differentiable solution path satisfying the Lagrange equations,

$$V' = f_x x^{*'} + f_\alpha.$$

The Lagrange FOC and the derivative of the binding constraint give

$$f_x = \lambda^* g_x, \quad g_x x^{*'} + g_\alpha = 0 \quad \Rightarrow \quad g_x x^{*'} = -g_\alpha.$$

Substituting,

$$\begin{aligned} V' &= \lambda^* g_x x^{*'} + f_\alpha \\ &= f_\alpha - \lambda^* g_\alpha \\ &= \boxed{\mathcal{L}_\alpha(x^*, \lambda^*, \alpha)}. \end{aligned}$$

The direct parameter effect remains, together with the parameter's effect on the binding constraint valued at the shadow price.

Inequality-constrained envelope

Consider a KKT problem in which the objective, constraints, and right-hand sides are differentiable functions of a scalar parameter α :

$$V(\alpha) = \max_x f(x, \alpha) \quad \text{s.t.} \quad g_j(x, \alpha) \leq b_j(\alpha).$$

With

$$\mathcal{L} = f + \sum_j \lambda_j (b_j - g_j),$$

along a differentiable solution with a **locally unchanged active set**,

$$\boxed{V'(\alpha) = \mathcal{L}_\alpha(x^*, \lambda^*, \alpha)},$$

The parameter derivative

$$\mathcal{L}_\alpha = f_\alpha(x^*, \alpha) + \sum_j \lambda_j^* (b'_j(\alpha) - g_{j,\alpha}(x^*, \alpha)).$$

- ◆ If only the objective depends on α , then $V'(\alpha) = f_\alpha(x^*, \alpha)$.
- ◆ If the parameter is a right-hand side b_j , then $\partial V / \partial b_j = \lambda_j^*$, recovering the shadow-price formula from KKT.

Envelope theorem: direct calculation and a shadow price

- (a) Let $V(\alpha) = \max_{x \in \mathbb{R}} \{\alpha x - \frac{1}{2}x^2\}$. Find $x^*(\alpha)$, compute $V'(\alpha)$ directly, and recover it from the envelope theorem.
- (b) For $b > 0$, let $V(b) = \max_{x,y>0} \{\ln x + \ln y : x + y = b\}$. Find $\lambda^*(b)$ and verify $V'(b) = \lambda^*(b)$.

Envelope theorem: direct calculation and a shadow price

For (a), the FOC gives $x^*(\alpha) = \alpha$, so

$$V(\alpha) = \frac{1}{2}\alpha^2, \quad V'(\alpha) = \alpha.$$

The envelope derivative is $f_\alpha(x^*, \alpha) = x^* = \alpha$. For (b), with $\mathcal{L} = \ln x + \ln y + \lambda(b - x - y)$, the FOCs give

$$x^* = y^* = \frac{b}{2}, \quad \lambda^* = \frac{2}{b}.$$

Since $V(b) = 2 \ln(b/2)$, direct differentiation gives $V'(b) = 2/b = \lambda^*(b)$.

Envelope theorem: a parameter inside the constraint

For $p, m > 0$, let

$$V(p) = \max_{x, y > 0} \{\ln x + \ln y : px + y \leq m\}.$$

- (a) Solve for x^* , y^* , and λ^* .
- (b) Use the envelope theorem to find V_p . Which term of $\mathcal{L}_\alpha$ on the preceding pages survives here?
- (c) Verify the answer by differentiating $V(p)$ directly.

Envelope theorem: a parameter inside the constraint

The objective is increasing in both goods, so the budget binds. With $\mathcal{L} = \ln x + \ln y + \lambda(m - px - y)$, the FOCs are

$$\frac{1}{x} = \lambda p, \quad \frac{1}{y} = \lambda, \quad \text{so} \quad px = \frac{1}{\lambda}, \quad y = \frac{1}{\lambda},$$

and $px + y = 2/\lambda = m$ gives

$$\lambda^* = \frac{2}{m}, \quad x^* = \frac{m}{2p}, \quad y^* = \frac{m}{2}.$$

Here p enters the constraint function $g(x, y, p) = px + y$, with $g_p = x$, while the objective and the right-hand side m are free of p . Only the term $-\lambda^* g_p$ of $\mathcal{L}_\alpha$ survives:

$$V_p = -\lambda^* x^* = -\frac{2}{m} \cdot \frac{m}{2p} = -\frac{1}{p}.$$

Directly, $V(p) = \ln \frac{m}{2p} + \ln \frac{m}{2} = 2 \ln \frac{m}{2} - \ln p$, so $V_p = -1/p$. Roy's identity below states this relation for a general utility function.

A binary optimizer jump and a kink in the value function

The binary choice again

Return to $x \in \{0, 1\}$ and $f(x, \theta) = x(\theta - k)$, so that $V(\theta) = \max\{0, \theta - k\}$. Away from $\theta = k$ the optimizer is constant on each side of k and the envelope formula holds:

$$V'(\theta) = f_{\theta}(x^*(\theta), \theta) = x^*(\theta) = \begin{cases} 0, & \theta < k, \\ 1, & \theta > k. \end{cases}$$

At $\theta = k$ the optimizer jumps from 0 to 1, the slopes $f_{\theta}(0, \theta) = 0$ and $f_{\theta}(1, \theta) = 1$ of the two curves differ, and V has no derivative. The kink comes from the differing slopes: a jump between curves whose slopes agree at the crossing leaves V differentiable there. Away from the jump the optimizer is constant, one case of the differentiable solution path the preceding pages assumed; at the jump there is no such path. The left and right derivatives are still the envelope values of the two competing choices.

KKT, parameter changes & envelope

Optional cumulative exercise: KKT, a parameter, and the envelope theorem

For $\alpha \geq 0$, consider

$$\sup_{x,y>0} \alpha \ln x + \ln y$$

subject to

$$2x + y \leq 6.$$

- For $\alpha > 0$, write the KKT conditions and solve for $x^*(\alpha)$ and $y^*(\alpha)$.
- At $\alpha = 0$, determine whether a maximizer exists and find the supremum.
- Let $V(\alpha)$ be the supremum. Use the envelope theorem to determine where V decreases and increases.
- Add $x \leq y$. Determine whether the solution changes at $\alpha = 1$ and $\alpha = 3$; find the revised solution when it changes.

KKT, parameter changes and envelope

Part (a): KKT system and solution

With multiplier $\lambda \geq 0$,

$$\mathcal{L} = \alpha \ln x + \ln y + \lambda(6 - 2x - y).$$

The KKT conditions are

$$\frac{\alpha}{x} = 2\lambda, \quad \frac{1}{y} = \lambda, \quad \lambda \geq 0, \quad 2x + y \leq 6, \quad \lambda(6 - 2x - y) = 0.$$

The objective is increasing in both variables, so the resource constraint binds: $2x + y = 6$. The first-order conditions imply

$$x = \frac{\alpha}{2\lambda}, \quad y = \frac{1}{\lambda}, \quad 2x + y = \frac{\alpha + 1}{\lambda} = 6.$$

Hence

$$x^*(\alpha) = \frac{3\alpha}{\alpha + 1}, \quad y^*(\alpha) = \frac{6}{\alpha + 1}, \quad \lambda^*(\alpha) = \frac{\alpha + 1}{6}.$$

KKT, parameter changes and envelope

Parts (b) and (c): the supremum at $\alpha = 0$ and the envelope derivative

At $\alpha = 0$, the objective is $\ln y$. Feasibility with $x > 0$ implies $y < 6$, so no maximizer exists and

$$V(0) = \sup_{x,y>0, 2x+y\leq 6} \ln y = \ln 6.$$

For $\alpha > 0$,

$$V(\alpha) = \ln 6 + \alpha \ln 3 + \alpha \ln \alpha - (\alpha + 1) \ln(\alpha + 1).$$

The envelope theorem gives

$$V'(\alpha) = \ln x^*(\alpha) = \ln \left(\frac{3\alpha}{\alpha + 1} \right).$$

Since $V(\alpha) \rightarrow \ln 6$ as $\alpha \downarrow 0$, V is continuous at zero. It decreases on $[0, 1/2]$ and increases on $[1/2, \infty)$.

KKT, parameter changes and envelope

Part (d): adding the constraint $x \leq y$

For $\alpha > 0$,

$$\frac{x^*(\alpha)}{y^*(\alpha)} = \frac{\alpha}{2}.$$

At $\alpha = 1$, $(x^*, y^*) = (3/2, 3)$, so $x \leq y$ is satisfied. At $\alpha = 3$, $(x^*, y^*) = (9/4, 3/2)$, so the added constraint changes the solution. The binding resource constraint gives $y = 6 - 2x$, and $x \leq y$ becomes $x \leq 2$. Strict concavity then places the revised solution at

$$(x^*, y^*) = (2, 2).$$

Roy's identity

Roy's identity from the value function

Define the indirect utility and expenditure functions by

$$v(p, w) = \max_{x \geq 0} \{u(x) : p \cdot x \leq w\}, \quad e(p, \bar{u}) = \min_{x \geq 0} \{p \cdot x : u(x) \geq \bar{u}\}.$$

If v is differentiable at (p, w) and $v_w(p, w) > 0$, Marshallian demand satisfies

$$x_i(p, w) = -\frac{v_{p_i}(p, w)}{v_w(p, w)}.$$

This is the envelope theorem applied to the consumer's value function.

Check with the Cobb-Douglas log utility of the worked example: $\tilde{v}(p, w) = \alpha \ln \frac{\alpha w}{(\alpha + \beta)p_1} + \beta \ln \frac{\beta w}{(\alpha + \beta)p_2}$ gives $\tilde{v}_{p_1} = -\alpha/p_1$ and $\tilde{v}_w = (\alpha + \beta)/w$, so $-\tilde{v}_{p_1}/\tilde{v}_w = \frac{\alpha}{\alpha + \beta} \frac{w}{p_1} = x_1^*$.

Video: [Economics in Many Lessons, Roy's identity](#).

Hotelling and Shephard

Envelope identities

Define the profit and cost functions by

$$\pi(p) = \max_{y \in Y} p \cdot y, \quad c(w, q) = \min_{x \geq 0} \{w \cdot x : F(x) \geq q\}.$$

At price vectors where π and $c(\cdot, q)$ are differentiable:

Hotelling	net supply from profit	$\frac{\partial \pi(p)}{\partial p_i} = y_i(p),$
Shephard	conditional factor demand from cost	$\frac{\partial c(w, q)}{\partial w_i} = x_i(w, q).$

With $c(q) = q^2/2$, the envelope exercise gave $q^*(p) = p$ and $\pi(p) = p^2/2$, so $\pi'(p) = p = q^*(p)$: Hotelling in one line. With $F(x) = \min\{x_1, x_2\}$, the cost function is $c(w, q) = (w_1 + w_2)q$, so $\partial c / \partial w_1 = q = x_1(w, q)$: Shephard in one line.

Videos: [Arizona Math Camp](#), [Hotelling's lemma](#); [Shephard's lemma](#).

Convex duality behind the value functions

Expenditure, cost, and profit

These value functions inherit curvature from pointwise extrema of linear functions of prices:

$e(p, \bar{u})$ and $c(w, q)$ are concave and homogeneous of degree one in their price vectors.

$\pi(p)$ is convex and homogeneous of degree one in netput prices.

Shephard's and Hotelling's identities connect price derivatives to conditional demands and net supplies.

Study strategy. When you cannot memorize all these properties in Micro I, derive the curvature and homogeneity from the value function definitions.

Further reading: Boyd, Ch. 7, §§7.4.3–7.5.6 (pp. 29–39); Ch. 13, §13.3.5 (pp. 24–25).

Profit and cost as envelopes of linear functions

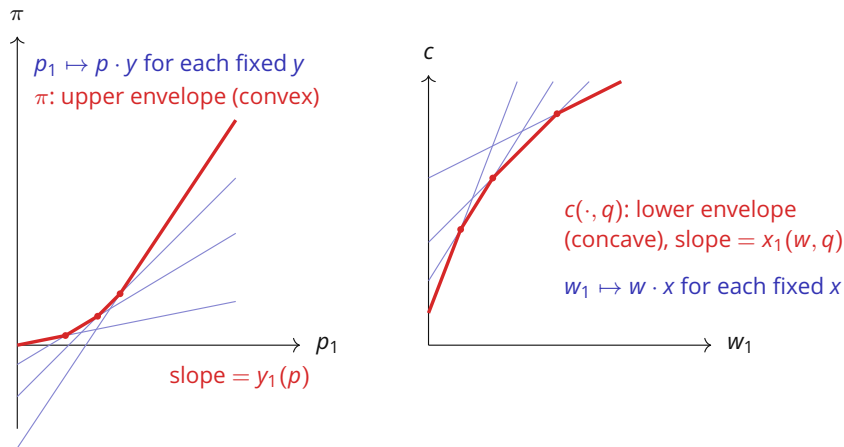


Figure 30 · Left: each fixed netput $y \in Y$ gives a line in p_1 ; π is their upper envelope, hence convex, with slope $y_1(p)$. Right: each fixed input bundle with $F(x) \geq q$ gives a line in w_1 ; $c(\cdot, q)$ is their lower envelope, hence concave, with slope $x_1(w, q)$.

Auction payment formula

Envelope characterization

In a single-object auction, type v wins with probability $q(v)$ and pays $p(v)$, so $U(v) = vq(v) - p(v)$.

When the direct mechanism is incentive compatible, U is absolutely continuous and

$$U'(v) = q(v) \text{ a.e.}, \quad U(v) = U(\underline{v}) + \int_{\underline{v}}^v q(t) dt.$$

Since $U(v) = vq(v) - p(v)$,

$$p(v) = vq(v) - U(\underline{v}) - \int_{\underline{v}}^v q(t) dt.$$

Once the allocation rule is fixed, payments are pinned down up to the lowest type's surplus. With $q(v) = v$ on $[0, 1]$ and $U(0) = 0$: $U(v) = \int_0^v t dt = v^2/2$ and $p(v) = v \cdot v - v^2/2 = v^2/2$.

Further reading: [Roughgarden, auction notes](#).

Section 7

Worked example

Cobb-Douglas utility maximization

Let $\alpha, \beta, p_1, p_2, w > 0$. Maximize $x^\alpha y^\beta$ subject to $p_1 x + p_2 y \leq w$ and $x, y > 0$. Monotonicity makes the budget bind. By the log trick, maximize $\alpha \ln x + \beta \ln y$ instead, with

$$\mathcal{L} = \alpha \ln x + \beta \ln y + \lambda(w - p_1 x - p_2 y).$$

The first-order conditions are

$$\frac{\alpha}{x} = \lambda p_1, \quad \frac{\beta}{y} = \lambda p_2, \quad p_1 x + p_2 y = w.$$

Thus

$$p_1 x = \frac{\alpha}{\lambda}, \quad p_2 y = \frac{\beta}{\lambda}.$$

Video: [Economics in Many Lessons, Utility Maximization with a Cobb-Douglas Utility Function.](#)

Cobb-Douglas utility maximization: solve the conditions

Adding the two expenditure equations and using the binding budget gives

$$w = p_1x + p_2y = \frac{\alpha + \beta}{\lambda} \Rightarrow \lambda^* = \frac{\alpha + \beta}{w}.$$

Substitution into the first-order conditions yields

$$x^* = \frac{\alpha}{\alpha + \beta} \frac{w}{p_1}, \quad y^* = \frac{\beta}{\alpha + \beta} \frac{w}{p_2}.$$

This one problem reuses three earlier chapters: Functions gives the strictly increasing log transformation, Differentiation gives the FOCs, and Convexity gives strict concavity of the log objective and convexity of the budget set. Hence the FOCs identify the unique global maximizer.

Further reading: [Border, Preference and Demand Examples, Sections 1–2.](#)

Expenditure shares and the multiplier

Expenditure shares

Demands spend fixed income shares $\alpha/(\alpha + \beta)$ and $\beta/(\alpha + \beta)$ on the two goods.

Multiplier

$$\lambda^* = \frac{\alpha + \beta}{w} = \tilde{v}_w.$$

It is the envelope derivative of the indirect log-value.

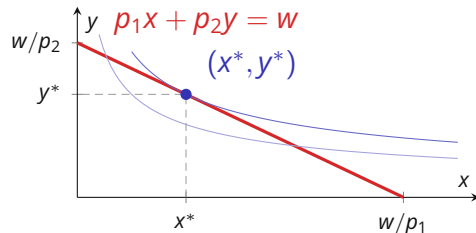


Figure 31 · Budget line and the tangent Cobb-Douglas indifference curve. With $\alpha : \beta = 1 : 2$, x^* is one third of w/p_1 and y^* two thirds of w/p_2 .

Cobb-Douglas utility with an upper bound on one good

Let $\alpha, \beta, p_x, p_y, w > 0$ and $0 < \bar{x} < w/p_x$. Solve

$$\max_{x,y>0} x^\alpha y^\beta \quad \text{s.t.} \quad p_x x + p_y y \leq w, \quad x \leq \bar{x}.$$

Use the log objective. Determine when the upper bound binds, find (x^*, y^*) in each case, and compute the multipliers on the budget and the upper bound.

Cobb-Douglas utility with an upper bound: choices

Without the upper bound, the demand for x is

$$x^u = \frac{\alpha}{\alpha + \beta} \frac{w}{p_x}.$$

Monotonicity makes the budget bind.
Therefore

$$x^* = \min\{x^u, \bar{x}\}, \quad y^* = \frac{w - p_x x^*}{p_y}.$$

If $\bar{x} > x^u$, the upper bound is slack. If $\bar{x} < x^u$, it binds and $(x^*, y^*) = (\bar{x}, (w - p_x \bar{x})/p_y)$. At $\bar{x} = x^u$, it binds with a zero multiplier.

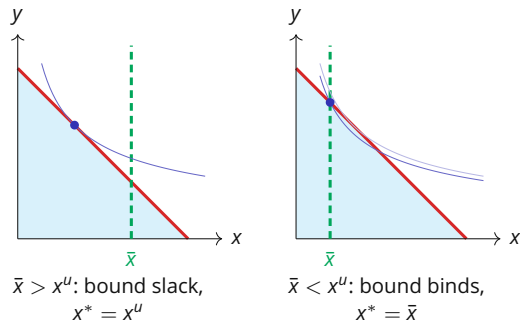


Figure 32 · The cap $x \leq \bar{x}$ (dashed) truncates the budget set. Left: $\bar{x}$ above the unconstrained demand x^u , the bound is slack. Right: $\bar{x}$ below x^u , the optimum sits on the cap where the indifference curve crosses the budget line.

Cobb-Douglas utility with an upper bound: multipliers

Use

$$\mathcal{L} = \alpha \ln x + \beta \ln y + \lambda(w - p_x x - p_y y) + \mu(\bar{x} - x).$$

When the upper bound is slack,

$$\lambda^* = \frac{\alpha + \beta}{w}, \quad \mu^* = 0.$$

When it binds,

$$\lambda^* = \frac{\beta}{w - p_x \bar{x}}, \quad \mu^* = \frac{\alpha}{\bar{x}} - \frac{\beta p_x}{w - p_x \bar{x}} \geq 0.$$

The last inequality is equivalent to $\bar{x} \leq x^u$. These are the shadow prices of wealth and the upper bound for the optimized log value.

Section 8

Further reading

Optimization references

Recommended references

Osborne, *Mathematical Methods for Economic Theory*, optimization sections [\[link\]](#).

Echenique, *Constrained Maximization* [\[link\]](#).

Border, *Miscellaneous Notes on Optimization Theory and Related Topics* [\[link\]](#).

Comparative statics and extensions

Selected references

Echenique, *Monotone Comparative Statics* [link];

Border, *Envelope Theorem* [link].

Border, *Single Crossing Properties* [link] and *Lattices and Supermodularity* [link].

Sarver, *Microeconomic Theory Lecture Notes*, Chapters 1–2 [link].

Original sources: Milgrom and Shannon (1994) [link]; Topkis (1998) [link].